

Quantification, Lived Experience, and Documentation AI: A Phenomenological Bioethics

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This paper critiques quantification overreach in AI bioethics: the tendency for metrics and models to move from representing phenomena to replacing them in ethical and practical judgment. We call this mechanism *ethical substitution*. The paper uses LLM-based EHR summarization for clinical handoffs as a stress test because documentation systems are relationship-shaping infrastructure, not neutral clerical tools. We argue that these systems can produce relational harm even when factually accurate through three mechanisms: compression of ambiguity, selective legibility, and authority shift. Conceptually, we combine phenomenological critique of mathematization with Canguilhem’s anomaly/abnormality distinction to show how statistical treatment of variation can collapse descriptive deviation into prescriptive abnormality. Constructively, we derive six design and governance requirements and a relationship risk map for evaluation beyond accuracy. The contribution is not methodological novelty in “phenomenology + AI,” but a traceable chain from mechanism diagnosis to actionable documentation-AI requirements.

1 Introduction

AI systems now mediate substantial portions of clinical work, from risk prediction and triage to increasingly documentation-centered workflows [22, 24, 29]. In parallel, AI ethics has accumulated an extensive vocabulary of principles, standards, and metrics. Yet reviewers of this literature have repeatedly noted a persistent mismatch between what can be measured and what should matter in practice: operationalization often advances faster than justification, and formal tractability often displaces social and relational complexity [11, 18, 20, 23, 31]. This paper addresses that mismatch in the subfield of AI bioethics.

Our central claim is that one recurring mechanism behind quantification overreach is *ethical substitution*: a quantified representation, produced to support judgment, gradually takes the place of the phenomenon it is meant to represent. In healthcare, this displacement is not merely epistemic; it is relational and institutional. Records and summaries shape credibility, determine what counts as clinically relevant, and influence whose account is treated as authoritative when decisions travel across teams and time. Once substitution occurs, the system can appear well-governed in metric terms while still eroding recognition and contestability.

We analyze this mechanism through a single hard case: LLM-based EHR summarization for clinical handoffs. The case is chosen because documentation artifacts are infrastructural. They are reused, copied forward, and recontextualized across clinical and administrative pathways. In that setting, three mechanism-level risks become salient: *compression* (uncertainty and ambiguity are converted into fluent closure), *selective legibility* (what is weakly encoded becomes weakly represented), and *authority shift* (summaries become default institutional truth through downstream reuse). These risks are conceptually distinct but operationally coupled.

The manuscript is written for three overlapping audiences. For conceptual-track readers in FAccT/AIES-style venues, it offers a mechanism-level account that narrows where and how quantification overreach occurs. For HCI/CSCW and design teams, it offers implementable requirements and a traceable rationale rather than abstract principles alone. For clinical-informatics researchers and governance stakeholders, it offers an evaluation artifact—the relationship risk map—that can be applied in review, deployment, and post-incident settings.

The genre of this contribution is a critical essay with an applied artifact. It is not a deployment study and does not claim empirical effect-size evidence. Its objective is to make an argument that is simultaneously philosophically

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explicit and operationally auditable: from critique, to mechanism, to design requirements, to evaluation consequences. The manuscript therefore prioritizes clarity of derivation over breadth of checklist.

The paper makes five linked contributions. First, it names ethical substitution as a distinct mechanism and positions it in relation to adjacent diagnoses, including epistemic calcification in healthcare AI [12]: calcification primarily describes a rigidified epistemic state, whereas substitution identifies the workflow process by which representation becomes stand-in. Second, it foregrounds Canguilhem’s anomaly/abnormality distinction as a resource for analyzing how AI/ML classification can collapse descriptive deviation into prescriptive disvaluation in health contexts. Third, it develops an applied analysis of LLM documentation infrastructure through the three mechanisms above—compression, selective legibility, and authority shift—to explain relational harm beyond factual error. Fourth, it derives six design and governance requirements from that mechanism analysis; the contribution is the derivation chain and bundling, not novelty of each requirement in isolation. Fifth, it introduces a relationship risk map as an evaluation artifact for documentation systems that need to be assessed beyond accuracy and throughput.

This positioning also sets clear boundaries. We do not claim methodological novelty in “phenomenology + AI,” a mature area of inquiry. We do not claim a new comprehensive ethical framework. We do not claim that each requirement is individually unprecedented. We do claim that the specific integration offered here—mechanism diagnosis, infrastructure stress test, traceable design implications, and evaluative artifact—is useful for addressing a practical gap repeatedly identified by both reviewers and practitioners: the gap between philosophical critique and actionable system governance.

Finally, the stance of the paper is explicitly bounded. It is not anti-measurement, anti-standardization, or anti-biomedical modeling. Quantification is indispensable in healthcare. The critique is about substitution and totalization: when quantified artifacts become the only admissible account of what is real in care relationships. The remainder of the paper develops this claim in eight steps: conceptual diagnosis, hard-case analysis, concise phenomenological corrective, design implications, evaluation strategy, and explicit tradeoff boundaries.

2 Quantification and Its Limits in Bioethics

Quantification in medicine has real and non-negotiable value. It enables comparability across sites, supports epidemiological coordination, and allows institutions to identify harmful variation at scale. Any critique that treats these achievements as incidental will fail both historically and practically. The question in this paper is therefore not whether healthcare should quantify, but where quantification becomes epistemically and ethically overextended.

In AI bioethics, overextension often appears when operationalization is treated as if it were closure: once a concern is converted into a score, benchmark, or optimization target, the representation can acquire authority beyond its scope. What starts as a tool for judgment can become a substitute for judgment. This is the transition we call ethical substitution.

2.1 Naturalistic framing and silent normativity

The tension is visible in familiar health-theory debates. In the biostatistical tradition, health is modeled through statistical regularity of function, and disease through departures from that regularity [2, 3]. This framework offers clarity and reproducibility, but its value-neutral self-description can obscure where normativity re-enters through reference-class construction, threshold choice, and interpretation of deviation.

Table 1. Three modes of quantification overreach in bioethics

Mode of overreach	How it appears technically legitimate	What is displaced	Downstream authority effect
Category closure	Stable labels and thresholds improve reproducibility and coordination.	Local context, evolving interpretation, and contested narratives.	“Class is case”: categories become hard to contest once reused.
False objectivity	Aggregate metrics appear perspective-free and institutionally neutral.	Voice asymmetries and interpretive labor required for judgment.	“Score is truth”: outputs treated as complete evidence.
Ethical substitution	Representation is operationally efficient and portable across workflows.	Phenomenon-level meaning in lived and relational contexts.	“Artifact is phenomenon”: summaries stand in for persons/cases.

AI systems inherit and amplify this problem because they depend on prior representational infrastructures. Datafication transforms heterogeneous encounters into machine-tractable records, codes, and templates [25, 26]. That transformation is never a neutral mirror of clinical reality. It is shaped by institutional incentives, workflow constraints, and schema design choices. Models then optimize against those representational decisions while often appearing to users as if they were directly grounded in the world itself.

Husserl’s critique of mathematization is relevant here not because it rejects formal science, but because it identifies a specific failure mode: method is mistaken for being, and technical adequacy is mistaken for epistemic completeness [14]. In healthcare AI, the analogous risk appears when model outputs are treated as if they exhaust the phenomenon they summarize.

2.2 Canguilhem and classification overreach

The strongest conceptual lever in this paper is Canguilhem’s anomaly/abnormality distinction [5, 6]. An anomaly marks descriptive variation relative to a distribution. Abnormality marks a prescriptive judgment that variation is deficient or disvaluable relative to a normative order of life. Conflating these two registers is a recurrent error in medical reasoning, and AI classification infrastructures make that error easier to institutionalize.

Predictive systems are built to partition variation under task constraints. That is not itself a defect. The problem begins when classificatory outputs are treated as if predictive utility automatically licenses normative conclusions about personhood, credibility, or acceptable forms of life. At that point, descriptive regularity is operationally converted into prescriptive authority, often without explicit deliberation. In practice, this can occur through seemingly routine steps: default label reuse, eligibility rules anchored to model categories, or downstream communication protocols that erase uncertainty and context.

This is exactly where Canguilhem adds something that standard accuracy/fairness discourse often cannot name. The issue is not merely skewed performance across groups; it is the production of normative force through classificatory infrastructure. In this sense, AI systems can silently transform “what is statistically common in data” into “what institutions treat as normal in care.”

2.3 Relation to adjacent critiques

This argument aligns with established critiques of quantification-centric AI ethics [11, 31]. It also aligns with phenomenology-informed concerns about decontextualization and erosion of first-person relevance in healthcare AI [7]. We therefore do not position ethical substitution as replacing these accounts.

The closest neighboring concept is epistemic calcification in healthcare AI [12]. The overlap is significant and should be explicit. Our proposed distinction is that calcification primarily describes a rigidified epistemic *state* (frameworks become fixed), while ethical substitution describes a workflow-level *mechanism* (representation becomes stand-in). The two can co-occur. In fact, substitution can be understood as one pathway through which calcification is stabilized in practice. The reason to keep the distinction is interventional: mechanism language identifies where authority shifts and where design or governance can interrupt that shift.

2.4 Register clarification: transcendental, not framework-competitive

Because “phenomenology + AI” is now a mature field, we clarify register. This paper does not claim novelty from using phenomenology as one ethical lens among others. It uses Husserlian phenomenology in a transcendental sense: to analyze conditions under which phenomena become intelligible and authoritative prior to downstream ethical codification [1, 13, 21, 33]. Ethical substitution is therefore treated as a structural failure in givenness and representation, not merely a procedural ethics defect.

This also explains why the paper does not market itself as a new normative framework. The goal is to diagnose a failure that any framework—deontological, consequentialist, virtue-theoretic, or justice-oriented—must still confront once quantified artifacts start replacing phenomena in institutional practice.

2.5 Post-phenomenology as neighbor field

Post-phenomenological mediation theory is an important neighboring tradition and not a target of displacement. Ihde’s work, including his later engagement with medicine, is highly relevant for artifact-level analysis of human–technology relations [16, 17]. Our focus differs in level of analysis: this paper addresses infrastructural mathematization and representational authority conditions that precede specific artifact interactions. This register distinction has been articulated within the literature itself [27, 30, 32]. Treating the traditions as complementary avoids a false rivalry and helps specify contribution boundaries.

2.6 Why bioethics and documentation are the decisive site

Bioethics concerns not only outcomes but admissibility of evidence, status of testimony, and obligations generated by mediated care relations. In this setting, quantification overreach is a governance issue before it is a purely technical one. Documentation infrastructures are where this governance issue becomes concrete: representational compression and normative force are braided together, and reused artifacts can organize institutional non-knowledge at scale.

For this reason, the remainder of the paper moves from conceptual diagnosis to a single infrastructure case—LLM-EHR summarization in handoffs—and then to design and evaluation consequences. The aim is to keep philosophical claims answerable to workflow realities and keep workflow proposals anchored in explicit mechanism analysis.

3 Why LLM Documentation Is a Hard Case

LLM-supported documentation has become a central site of healthcare AI adoption because it promises immediate organizational returns: reduced clerical burden, faster handoffs, and more legible summaries across teams [19, 28]. These gains are credible and often welcome in overburdened settings. But the ethical significance of documentation AI cannot be captured by productivity framing alone. Documentation is not a neutral byproduct of care. It is one of the infrastructures through which care is coordinated, justified, audited, and remembered.

This infrastructural role matters for at least three reasons. First, documentation artifacts travel: what is written in one encounter can structure subsequent interpretation by actors who were not present in that encounter. Second, documentation artifacts are multifunctional: they are used for handoff, billing, quality review, legal defense, and increasingly patient-facing communication. Third, documentation artifacts are recursive: once copied forward, they shape what becomes salient in the next round of note-taking and model generation. In this sense, documentation systems do not merely capture clinical reality; they help produce the reality institutions can act upon [4, 25, 26].

From this perspective, LLM summarization is a hard case because it accelerates existing representational dynamics and adds new portability and fluency effects. We focus on clinical handoff because it is both time-pressured and relationally consequential: small representational choices can alter trust, credibility, and accountability across teams and with patients.

3.1 A workflow-level stress test

Consider a plausible longitudinal workflow. During a shift transition, a clinician receives a model-generated summary synthesized from prior notes, structured fields, and recent updates. The summary is accepted with modest edits because time is limited and the artifact is coherent. Parts of that text are then reused in referral communication, discharge instructions, and patient portal messaging. By the time the representation reaches patient-facing contexts, it may be several transformations removed from the originating encounter, yet still treated as authoritative due to institutional provenance.

This chain illustrates why model-level quality indicators are insufficient for governance. Even when local output looks acceptable, representational effects accumulate downstream. The question is not simply whether a sentence is accurate in isolation, but what authority the sentence acquires through reuse, and what kinds of correction remain possible once it has propagated.

3.2 Three coupled risk mechanisms

The manuscript names three risk mechanisms that repeatedly appear in this workflow.

Compression. Clinical situations are often interpretively open: symptoms are ambiguous, hypotheses compete, and relevant context is distributed across narrative fragments. LLM summaries can convert this openness into fluent closure. Compression is therefore not only shortening; it is loss of epistemic structure—especially uncertainty, disagreement, and temporality.

Selective legibility. Models amplify what is consistently encoded and underrepresent what is weakly documented or schema-misaligned. Social determinants, caregiver observations, stigma-sensitive context, and contested self-accounts can become informationally peripheral even when clinically central. What is not easily represented risks being treated as if it were not there.

Authority shift. Once summaries are copied forward across artifacts and uses, they can acquire authority disproportionate to their epistemic status. Downstream actors often encounter the summary but not the originating encounter. In such settings, the artifact can become “the case” for practical purposes, especially under time pressure.

These mechanisms are analytically distinct but operationally coupled. Compression makes selective legibility harder to detect; selective legibility makes authority shift more harmful; authority shift, once established, incentivizes future compression. Together they explain how relational harms can emerge even when factual error rates appear low.

3.3 Relational harm beyond factual error

A summary can be factually plausible yet still produce misrecognition. It may preserve diagnosis while erasing uncertainty about causality. It may preserve treatment chronology while omitting caregiving constraints. It may preserve coded history while narrowing whose voice appears authoritative in downstream communication. In each case, the harm is not only informational. It concerns how people are positioned in care relationships and what kinds of contestation remain institutionally possible [7, 9].

This is why documentation AI should be governed as relationship-shaping infrastructure rather than as an isolated NLP component. Trust, recognition, and repair are not optional “soft” outcomes. They are constitutive of clinical validity when records mediate decisions across time and actors.

4 Phenomenological Corrective in Brief

The corrective in this paper is intentionally concise. We do not provide a full reconstruction of phenomenological method. We use a small conceptual toolkit to constrain overreach and to ground design/evaluation implications.

The first concept is **lifeworld**. Formal representations derive sense from lived practices and cannot replace that ground without distortion [14]. The second is **intersubjectivity**. Objectivity is not achieved by eliminating perspective; it is achieved through accountable coordination across perspectives. The third is **record-making as mediation**. Classification and documentation formats are productive infrastructures that shape what can count as a case [4, 25, 26]. The fourth is **renewal**: recurring critical reassessment of inherited representational practices and their normative consequences [15].

Taken together, these concepts yield a practical criterion: quantification remains legitimate when outputs stay revisable, contestable, and context-accountable within care workflows. Quantification overreaches when workflows convert outputs into unquestioned stand-ins for persons and situations.

In this way, phenomenology functions here as a grounding discipline for governance design, not as a substitute for empirical evaluation. It clarifies what must remain visible if documentation systems are to support coordination without displacing the relational and interpretive conditions of care.

5 Design and Governance Implications

If ethical substitution is the mechanism-level risk, the practical question becomes how to configure documentation workflows so summaries remain revisable aids rather than authority endpoints. This section translates the critique into six requirements. We emphasize at the outset that these requirements are not offered as a new universal framework. Several overlap with existing responsible-AI guidance when viewed in isolation. The contribution is the derivation logic: why these requirements, in this bundle, are specifically warranted for documentation infrastructure where compression, selective legibility, and authority shift are coupled.

To keep this derivation explicit, each requirement is linked to a mechanism-level failure and a governance objective. The relevant unit of intervention is the socio-technical workflow, not model architecture alone.

5.1 Draftness as an authorship safeguard

First, summaries should be treated operationally as drafts. This requires more than a textual disclaimer. It requires interface and process choices that make revision expected, visible, and accountable: tracked edits, explicit acceptance events, and preserved evidence of uncertainty when edits are made under time pressure.

The governance objective is to preserve authorship responsibility. Without draftness, compression pressure plus workflow speed can normalize “accept as-is” behavior and move authority to model output by default. Draftness keeps interpretive labor visible and prevents the appearance that the generated artifact is already institutionally final.

5.2 Provenance with omission signaling

Second, summaries should expose provenance in a way routine users can interpret, not only specialists debugging model behavior. Provenance should identify source windows and input scope, but also declare meaningful absences: what classes of context were unavailable, excluded, or weakly represented.

The governance objective is to prevent selective legibility from being misread as completeness. In documentation settings, users frequently interpret silence as non-existence. Omission signaling interrupts that inference by making partiality explicit, thereby reducing substitution pressure.

5.3 Multi-voice representation instead of narrative flattening

Third, summaries should preserve voice distinctions among patient account, caregiver context, and clinician interpretation. This can be implemented through structural segmentation, attribution tags, or role-specific views. The specific UI pattern may vary; the principle does not.

The governance objective is to resist false objectivism. Authority shift intensifies when heterogeneous claims are fused into a single fluent narrative that appears perspective-free. Multi-voice structure keeps disagreement intelligible and protects contestability in downstream contexts where original participants are absent.

5.4 Uncertainty and disagreement as first-class states

Fourth, systems should support unresolved uncertainty and active disagreement across handoffs. This includes the ability to carry forward competing hypotheses and to mark contested claims without forcing premature closure for workflow completion.

The governance objective is to counter compression’s most harmful effect: closure masquerading as certainty. In many cases, good care requires explicitly preserving “not yet resolved” rather than converting interpretive openness into finalized prose.

5.5 Repair loops with downstream propagation

Fifth, correction pathways must be routine and infrastructural. Clinicians should be able to amend summaries with clear visibility. Where institutionally feasible, patient and caregiver correction pathways should exist as well, at least through structured addenda or flag-and-review processes. Most importantly, repairs should propagate to downstream artifacts rather than remaining local.

Table 2. Traceability from mechanism diagnosis to derived requirement

Risk mechanism	Typical workflow failure	Derived requirement(s)	Status in standard frameworks
Compression	Uncertainty is converted into fluent closure; unresolved issues disappear during transfer.	Draftness by default; uncertainty/disagreement as first-class.	Partly addressed (human oversight common; uncertainty carry-forward less explicit).
Selective legibility	Weakly encoded context becomes invisible and is interpreted as clinically irrelevant.	Provenance with omission signaling; multi-voice representation.	Partly addressed (transparency common; omission signaling and voice structure uncommon).
Authority shift	Reused summaries become institutional ground truth beyond initial context and purpose.	Repair loops with propagation; purpose separation and reuse governance.	Rarely explicit as coupled governance requirement.

The governance objective is to prevent persistent substitution after error detection. If authority has already shifted through copy-forward reuse, local correction without propagation leaves the harmful representation operationally active elsewhere.

5.6 Purpose separation and reuse governance

Sixth, organizations should explicitly separate summary purposes and govern cross-context reuse. A handoff summary optimized for rapid clinical transfer should not silently become a billing, triage, and patient-facing artifact without recontextualization.

The governance objective is to bound authority transfer. Reuse is often where substitution becomes institutionally entrenched: what was “good enough” for one purpose acquires coercive force in another because it arrives with inherited legitimacy.

5.7 Traceability from mechanism to requirements

Table 2 addresses a likely objection directly: that these requirements are simply standard principles in new language. The table therefore includes a final column indicating whether each requirement is typically explicit in standard health-care AI guidance. This makes novelty claims auditable and modest: the chain and bundling are central, not isolated requirement originality.

6 Evaluation Beyond Accuracy

If substitution risk is relational and infrastructural, then evaluation confined to model-output quality will under-detect failure. Documentation systems should be assessed as relationship infrastructures with technical components, not as technical components with optional relationship effects.

We propose an evaluation bundle with four coupled layers. The quantitative layer tracks omission frequency, uncertainty suppression, override/edit patterns, and downstream propagation of early representational errors. The workflow

Table 3. Relationship risk map (worked examples for documentation AI)

Relationship	Artifact	Primary mechanism	Likely relational harm	Repair path-way	Evaluation question
Clinician–clinician	Handoff summary	Compression	Unresolved differentials collapse into one narrative, reducing coordination under ambiguity.	Uncertainty addendum plus contested-items field.	How often are unresolved items preserved versus collapsed at handoff?
Patient–clinician	Portal or discharge reuse	Selective legibility	Lived context is omitted, producing misrecognition and trust erosion.	Patient flag-and-amend with visible clinician response.	What share of patient-reported omissions are acknowledged and integrated?
Patient–institution	Secondary reuse across billing, triage, and quality	Authority shift	Partial narratives become difficult-to-contest institutional defaults.	Purpose-gated reuse with propagated correction logs.	When a summary is corrected, where else is the correction reflected and how fast?

layer examines adoption under realistic time pressure, distribution of revision burden across roles, and correction latency. The qualitative layer assesses recognition, trust, and contestability through stakeholder interviews and targeted case reviews. The governance layer checks whether repair actions remain visible and effective after artifact reuse.

To operationalize these layers, we introduce a relationship risk map with worked examples (Table 3). The map is intentionally practical: it links relationship-artifact pairs to mechanism-level risks, likely harms, repair pathways, and concrete evaluative questions.

The risk map is intended to be used in design reviews, procurement assessments, and post-incident retrospectives. Its main advantage is diagnostic orientation: it shifts the governing question from “Is the output good?” to “What kind of authority does this artifact exercise in this relationship, and what happens when it is contested?”

This evaluation stance also clarifies what counts as success. A system can achieve strong benchmark performance and still fail if it systematically suppresses uncertainty, narrows voice, or makes repair practically inaccessible. For documentation AI, these are not peripheral harms. They are validity conditions for responsible deployment.

7 Tradeoffs, Boundaries, and Related Positions

The argument developed in this paper can be misunderstood in two opposite ways. One misreading is that it is anti-quantification and therefore impractical for contemporary healthcare. The other is that it simply repackages familiar responsible-AI principles in philosophical language. This section addresses both concerns by stating explicit tradeoffs, clarifying scope limits, and positioning the contribution relative to adjacent work.

7.1 Why this is not anti-measurement

Quantification is integral to clinical practice and health governance. Standardized records support coordination across institutions; epidemiological indicators support intervention planning; quantitative tracking can reveal inequities that anecdotal reporting may miss. In this sense, measurement is often a precondition for accountability rather than its enemy.

Historical and contemporary examples underscore this point. Quantified public-health practices have enabled large-scale gains under specific institutional conditions [10]. Feminist and justice-oriented data projects have also shown

that counting, when reflexively governed, can expose violence and institutional neglect that would otherwise remain politically invisible [8]. A credible critique of quantification must preserve these possibilities.

Our claim is therefore bounded: the target is not quantification but totalization. Problems emerge when quantified artifacts are granted authority as if they exhausted clinically and ethically relevant reality. In documentation infrastructures, that overreach is most visible when artifacts become difficult to contest after reuse.

7.2 What this paper adds beyond generic principles

A fair objection is that many of the proposed requirements resemble standard responsible-AI language: oversight, transparency, contestability, governance. We agree at the level of vocabulary overlap. The distinction is at the level of derivation and coupling.

First, the requirements in this manuscript are not presented as independently novel controls. They are derived from a specific mechanism diagnosis in a specific infrastructure context. Second, the requirements are coupled rather than modular: for example, provenance without omission signaling leaves selective legibility intact; repair without propagation leaves authority shift intact. Third, the evaluation artifact (relationship risk map) is designed to preserve this coupling in deployment decisions.

In other words, the contribution is not “a better checklist.” It is a mechanism-to-governance chain that explains why some controls that look adequate at the model layer are insufficient at the workflow layer.

7.3 Complementarity with adjacent critiques

The manuscript is intentionally complementary to existing critical AI ethics literature. It extends critiques of quantification overreach and epistemic authority [11, 31] by specifying documentation-specific pathways through which authority transfer becomes institutionalized. It is also complementary to phenomenology-informed critiques of deontologicalization in healthcare AI [7].

The nearest conceptual neighbor is epistemic calcification [12]. We do not reject this framing; we differentiate emphasis. Calcification names rigidity in epistemic frameworks. Ethical substitution names the operational process by which representations become stand-ins in concrete workflows. The distinction is maintained because intervention points differ: calcification diagnoses structural closure, while substitution helps specify where in the chain repair, contestability, and purpose governance should be introduced.

7.4 Scenario limits and empirical next steps

This paper remains scenario-based and should be read accordingly. It offers mechanism analysis and governance implications grounded in plausible documentation workflows, not causal effect estimates across deployed systems. That limitation is substantive, not cosmetic.

Three empirical next steps follow directly. The first is deployment-grounded validation of the three mechanisms across heterogeneous clinical settings (e.g., ED, inpatient transfer, outpatient chronic-care follow-up). The second is comparative testing of requirement bundles to determine which combinations most effectively reduce substitution risk under real constraints. The third is longitudinal study of correction propagation to evaluate whether repair pathways remain effective once summaries are reused beyond initial contexts.

These next steps do not weaken the conceptual contribution; they locate its proper use. The present manuscript is designed to make empirical programs better targeted by clarifying what should be measured and why.

7.5 Boundary condition: bounded pluralism

Bounded pluralism is the paper’s explicit boundary condition. It means that responsible documentation AI requires both quantitative discipline and interpretive accountability, and that neither can be treated as a complete replacement for the other. This is not offered as a novel doctrine. It is offered as a practical guardrail against overclaiming.

Under this guardrail, quantified outputs remain legitimate as revisable aids embedded in accountable relationships. Where systems suppress uncertainty, collapse voice, or block repair propagation, their outputs should not receive automatic institutional authority regardless of benchmark quality. The practical governance question is therefore not “Does the system quantify?” but “How is quantified output positioned in relations of care, and what happens when it is contested?”

8 Conclusion

This paper argued that a central risk in AI bioethics is ethical substitution: the point at which a quantified representation stops serving as an aid and starts functioning as the phenomenon itself in practical judgment. We analyzed that risk in a single infrastructure case—LLM-based documentation support for clinical handoffs—and showed why harms can emerge even when local output quality appears acceptable.

The main value of the paper is not a standalone philosophical diagnosis and not a standalone design checklist. It is the traceable chain connecting them: mechanism diagnosis (substitution), workflow mechanisms (compression, selective legibility, authority shift), derived governance requirements, and an evaluation artifact (relationship risk map) that can be applied beyond this manuscript. This chain is what allows a conceptual argument to become reviewable in design and deployment settings.

The paper also set explicit boundaries. It does not claim novelty from applying phenomenology to AI in general. It does not claim each requirement is individually unprecedented. It does not claim to replace empirical evaluation. It does claim that documentation systems should be governed as relationship-shaping infrastructures and that model-centered evaluation is insufficient where representational authority travels across care pathways.

Finally, the argument remains bounded with respect to quantification itself. Measurement and standardization are indispensable in healthcare. The critique is directed at totalization: when quantified artifacts become difficult to contest and are treated as exhaustive accounts of persons and cases. A phenomenological bioethics, on this view, contributes a practical discipline of limits—keeping representation accountable to lived context, preserving uncertainty where warranted, and making repair possible where substitution has already occurred.

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